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Advanced Fluid Modeling of Nanoparticles in Heterogeneous Rocks: Paving the Path for Sustainable Carbonate Reservoir Development

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Abstract

Carbonate reservoirs, which contain a significant portion of the world's oil reserves, present unique challenges due to their complex pore structures and oil-wet characteristics, which hinder efficient oil extraction. The utilization of nanoparticles in improved and enhanced oil recovery (IOR/EOR) has emerged as a promising approach to address these challenges. Additionally, the modeling of nanoparticle behavior in carbonate rocks remains an area with significant gaps in understanding. This paper aims to develop and validate an advanced fluid modeling framework that integrates nanoparticle transport and interaction mechanisms in carbonate rock reservoirs under reservoir conditions. The primary objective is to identify and quantify critical parameters such as adsorption, wettability alteration, and transport dynamics—that influence the efficiency of nanoparticle-assisted oil recovery. The study involves complex mechanisms to understand these parameters and unlock the potential of nanoparticles in improving oil recovery.

In this study, a comprehensive experimental and numerical modeling approach was employed to evaluate the performance of nanoparticles in coreflooding experiments. Laboratory tests involved fluid and rock characterization, adsorption rate, the injection of nanoparticle-enhanced fluids into carbonate core samples, with measurements taken for differential pressure, flow rates, effluent composition, adsorption. The experimental setup involved injecting the fluids into carbonate core samples under reservoir conditions, while monitoring pressure, flow rates, and effluent volume. The developed model incorporates key physical and physico-chemical processes, including wettability and adsorption measurements.

The experimental coreflooding tests, which included rocks-fluid characterization and nanofluid injection, were successfully validated through a robust modeling framework. The results demonstrate that adsorption plays a crucial role in the efficiency of the oil displacement process. Adsorption of nanofluids onto the rock surface was investigated to significantly alter the wettability of the carbonate matrix, enhancing oil displacement and recovery. The integration of experimental insights with predictive modeling enables a deeper understanding of nanoparticle transport and interaction within complex carbonate matrices.

The findings provide valuable insights into the design of next-generation IOR technologies, presenting a predictive model for a cost-effective and environmentally friendly solution to maximize hydrocarbon recovery and ultimately contributing to the sustainable development of carbonate oil fields.

Introduction

The carbonate reservoirs present unique challenges due to their inherent heterogeneity, complex pore structures, and variable wettability characteristics (Wilson, 2017). Unlike clastic reservoirs, carbonate formations often exhibit bimodal porosity systems, including interparticle pores, vugs, and fractures, which complicate fluid flow modeling and recovery predictions (Wilson, 2017). As the energy industry transitions toward more sustainable practices, enhancing recovery from carbonate reservoirs while minimizing environmental impact has become a strategic priority (Narr et al., 2006).

Nanotechnology has emerged as a promising solution to improve oil recovery and reservoir characterization. Nanoparticles (NPs), due to their ultra-small size and tunable surface properties, can alter interfacial tension, modify wettability, and improve mobility control in porous media (Zhang et al., 2017). Recent studies have demonstrated that engineered nanoparticles can penetrate deep into reservoir matrices, interact with rock surfaces, and enhance oil displacement efficiency. Moreover, the role of nanoparticles in mitigating formation damage and enhancing CO₂-EOR (Enhanced Oil Recovery) processes has gained attention. Studies have shown that silica and metal oxide nanoparticles can stabilize CO₂ foams, reduce capillary trapping, and improve miscibility in carbonate reservoirs (Al-Shargabi, et al., 2022). These effects are particularly beneficial in reservoirs with high water saturation and low permeability zones, where conventional EOR methods struggle.

However, the integration of nanoparticles into fluid modeling frameworks remains underdeveloped, particularly in the context of heterogeneous carbonate systems. Advanced fluid modeling techniques are essential for capturing the complex interactions between nanoparticles, reservoir fluids, and rock matrices. Incorporating nanoparticle behavior into these models requires a multi-scale understanding of transport mechanisms, adsorption dynamics, and phase behavior alterations. For instance, nanoparticle-induced wettability alteration has been shown to shift relative permeability curves and improve sweep efficiency in fractured carbonate cores (Tajmiri and Ehsani, 2017).

Hendraningrat et al. (2023) developed of a novel, robust three-dimensional reservoir simulation model tailored for field-scale application of nanoparticles in improved and enhanced oil recovery (IOR/EOR). Traditional commercial simulators lack the capability to accurately represent nanoparticle phase behaviour and transport mechanisms. To address this, the authors designed advanced algorithms that incorporate physicochemical interactions between nanoparticles, porous media, and reservoir fluids—capturing phenomena such as aggregation, splitting, and solid-phase deposition. The model was validated using multiple laboratory coreflood experiments in sandstone reservoirs and achieved a quality index match of approximately 95%. It was further refined by training with multiple published datasets involving various nanoparticle types, chemical additives, water hardness, salinity ranges, rock permeability, and oil viscosity. Critical parameters such as aggregation rate, adsorption, and retention factor were identified as essential for upscaling to field-scale simulations. The result is a simulation framework expect to able of accurately predicting nanoparticle-assisted recovery performance across diverse reservoir conditions. This work establishes a new reference for integrating nanotechnology into reservoir modelling, especially sandstone reservoir and offers a practical tool for optimizing EOR strategies using nanoparticles at the field scale.

Meanwhile, Jumiati et al. (2024) investigated new finding on the dynamic adsorption behaviour of nanosilica particles and their impact on enhanced oil recovery (EOR) in carbonate reservoirs. Two types of nanosilica, LHP (lipophobic and hydrophilic polysilicones) and HLP (hydrophobic and lipophilic polysilicones), were tested on carbonate rocks using Sumatran crude oil. A series of experiments were

conducted, including wettability alteration, interfacial tension, fluid stability, zeta potential, UV-vis spectrophotometric adsorption, and coreflood injection scenarios. The results showed that LHP nanosilica significantly improved oil recovery in neutral-wet carbonate rocks, achieving recovery rates between 9% and 45%. A direct correlation was investigated between nanosilica adsorption (0.06–0.15 mg/g) and oil recovery, indicating that higher adsorption led to greater recovery efficiency.

Since there is a gap in carbonate rock modeling and new finding on adsorption, the objective of this study is to develop and validate an advanced fluid modeling framework that integrates nanoparticle transport and interaction mechanisms in carbonate rock reservoirs under reservoir conditions. Therefore, critical parameters can be identified such as adsorption, wettability alteration, and transport dynamics that influence the efficiency of nanoparticle-assisted oil recovery. The study involves complex mechanisms to understand these parameters and unlock the potential of nanoparticles in improving oil recovery. The model leverages laboratory data, including nanoparticle dispersion, adsorption isotherms, and coreflooding experiments, to calibrate simulation parameters. It also incorporates pore-scale imaging and digital rock physics to capture the spatial variability of carbonate rock properties.

Material

The dead crude oil sample employed is classified as light oil (viscosity ~ 5 cP), with an API gravity of 40° , sourced from a hydrocarbon reservoir located in Sumatra, Indonesia. The reservoir temperature is recorded at approximately 65°C , which is representative of typical subsurface thermal conditions in the region. A detailed crude oil compositional assessment was performed using the SARA fractionation method, which categorizes crude oil into four primary components: Saturates, Aromatics, Resins, and Asphaltenes. The analysis revealed that the oil sample is predominantly composed of resins, accounting for 70% of the total composition, followed by saturates at 21%, aromatics at 8%, and a minor fraction of asphaltenes at 0.22%. The synthetic brine employed during the experimental procedures was laboratory-prepared to replicate actual field conditions, with a salinity of 5,000 ppm and a measured density of 1.0876 g/cm^3 .

Nanoparticles (NPs) used in the study were characterized using both Scanning Electron Microscopy (SEM) and Transmission Electron Microscopy (TEM), which revealed particle sizes ranging from 9 to 17 nanometers. Mineralogical analysis of the NPs was conducted via X-ray fluorescence (XRF), as illustrated in Figure 1, confirming that the nanoparticles are composed entirely of silica. For rock-fluid interaction studies, carbonate core samples were selected with average permeability around 2 mD. Table 1 presents the petrophysical properties and mineralogical composition of the cores used in coreflooding experiments, indicating that both samples are predominantly composed of carbonate minerals.

Table 1—Core properties and mineralogy analysis. They are carbonate cores for this experiments.

Parameter	Unit	Carbonate Core
Length	cm	5.55
Diameter	cm	2.9
Pore Volume	cm^3	11.3
Porosity	fraction	0.11
Mineralogy Analysis		
Ca	wt. %	85.77
Si	wt. %	7.81
Al	wt. %	3.97
Fe	wt. %	1.98
K	wt. %	0.16

Parameter	Unit	Carbonate Core
Ti	wt. %	0.31

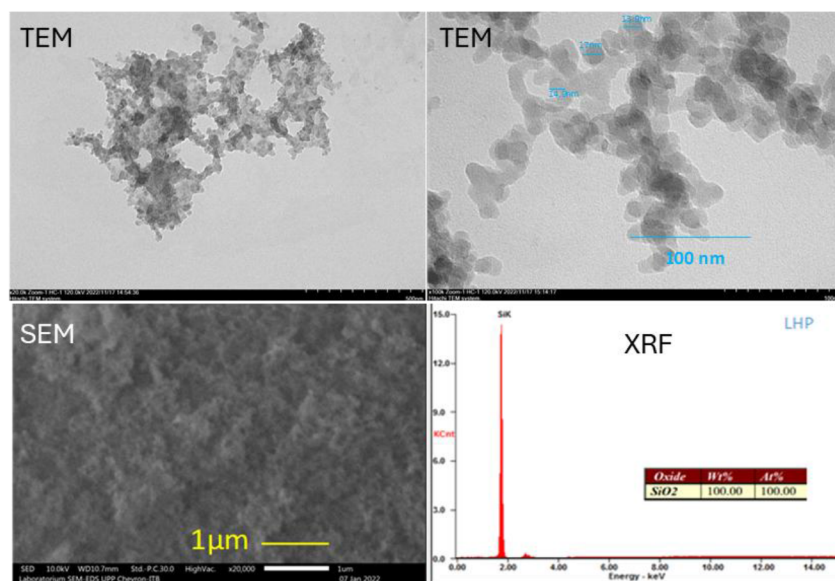


Figure 1—Characterization of Employed Nanoparticles using SEM, TEM, and XRF. Particle sizes range from 9 to 17 nm and are composed entirely of silica. These properties are critical for understanding nanoparticle behavior in porous media.

Methodology

A comprehensive laboratory and numerical studies validation were carried out using CMG-STARs with a newly developed process wizard is named as PETRONAS AFM (Hendraningrat et al., 2023). The general algorithms were developed by incorporating important physic-chemical interactions that exist across nanoparticles along with the porous media and fluid: phase behaviour and flow characteristic of nanoparticles that include aggregation, splitting and solid phase deposition. A new reaction stoichiometry was introduced to capture the aggregation process. The new algorithm was also incorporated to describe disproportionate permeability alteration and adsorption of nanoparticles, aqueous phase viscosities effect, interfacial tension reduction, and rock wettability alteration. The newly developed process wizard has demonstrated to predict sandstone reservoir that involved various types of nanoparticles, different chemical additives, hardness of water, wide range of water salinity and rock permeability and oil viscosity from ambient to reservoir temperature.

Laboratory Experiments

Nanofluids were formulated by dispersing nanoparticles (NPs) at a concentration of 0.1 wt.% into an aqueous medium, with the addition of 1 wt.% polyvinylpyrrolidone (PVP) serving as a stabilizing agent to enhance colloidal stability and prevent agglomeration. A synthetic brine with a salinity of 5,000 ppm was subsequently introduced into the nanoparticle-PVP mixture. The resulting solution was subjected to mechanical stirring at a rotational speed of 1,000 rpm for a duration of one hour under ambient laboratory conditions to ensure homogeneity. Following the stirring process, the nanofluid was further treated using ultrasonic agitation in a batch sonicator for an additional hour to promote uniform dispersion and enhance stability at the nanoscale. Table 2 provides a comprehensive summary of the experimental procedures conducted, encompassing both fluid–fluid and fluid–rock interaction tests, which were designed to evaluate the performance and compatibility of the nanofluids under reservoir-relevant conditions.

Table 2—Summary of Experimental activities of this study.

No.	Experiment	Description	Instrument
1	Fluid Characterization	Classify the components of crude oil into four distinct fractions: Saturates, Aromatics, Resins, and Asphaltenes.	SARA fractionation method
2	Characterization of Nanoparticles	Identify the size of nanoparticles.	Transmission Electron Microscope, Scanning Electron Microscope, X-ray fluorescence (XRF)
3	Core mineralogy analysis	Identify minerals in the core	X-ray fluorescence (XRF)
3	Thermal stability	Test to investigate the stability of nanofluid at reservoir temperature. Good NPs will have no physical and chemical changes in room and reservoir temperature.	Heating cabinet at reservoir temperature
4	Rock Wettability	Measure the wettability state of the core plugs through wettability index	Amott Cell
5	Interfacial tension (IFT)	Interfacial tension (IFT) measurement is conducted on the formulation at reservoir temperature to investigate the IFT reduction (if any) from the interaction of the product to the Oilfield.	Spinning drop video tensiometer model TX 500D
6	Adsorption nanoparticles	A spectrophotometer is an instrument that consists of a spectrometer and a photometer. The spectrometer generates light from the spectrum at a specific wavelength, and the photometer is a device that measures the intensity of light that is transmitted or absorbed.	Spectrophotometer UV-VIS

The thermal stability of the synthesized nanofluids was evaluated via sedimentation analysis conducted under reservoir-temperature conditions. The nanofluid samples were stored at heating cabinet, and visual inspection confirmed that the dispersion remained stable without noticeable sedimentation or phase separation for a duration exceeding 30 days.

Coreflooding experiments were subsequently performed to assess the enhanced oil recovery (EOR) potential of the nanofluid under simulated reservoir conditions. The tests were conducted within a pressure range of 200–300 psia and at a constant temperature of 65 °C, with a confining pressure maintained between 250–300 psia to replicate in-situ stress conditions. The experimental setup included a precision-controlled constant flow rate pump, calibrated to deliver fluid at a rate of 0.3 cc/min. Effluent volumes were collected and quantified throughout the flooding sequence, and these measurements were utilized to calculate adsorption, fluid saturation profiles and determine the recovery factor (RF).

The coreflooding protocol followed a sequential injection strategy, beginning with conventional waterflooding (WF1) to establish baseline recovery. This was followed by nanofluid injection (nanoflooding) to evaluate incremental oil displacement, and finally, a chased waterflood (WF2) was applied to assess post-nanofluid sweep efficiency. The residual resistance factor (RRF) or flow performance test on core plugs were calculated from differential pressure data to show the change in permeability after the injection treatment. The comparative analysis of recovery factors across these stages provided insights into the synergistic effects of nanoparticle-assisted EOR mechanisms.

Modeling Development

A coreflood simulation model was developed and subsequently refined using experimental data obtained from two distinct core samples. The numerical model was constructed in one-dimensional (1D) format, consisting of 100 uniformly distributed grid blocks. The total pore volume of the model was calibrated to replicate the actual pore volume of the physical core samples, ensuring consistency between laboratory and simulation environments. To conform with reservoir simulation standards, the core dimensions were converted from laboratory scale to field units, facilitating accurate scaling and interpretation of results.

Key rock properties—including porosity, absolute permeability, and initial oil saturation—were incorporated directly from laboratory measurements to preserve geological fidelity. History matching was performed by adjusting selected parameters within acceptable physical ranges to achieve alignment between simulated and observed data during nanofluid injection experiments. The primary objective functions used to evaluate model performance were differential pressure (DP) across the core and cumulative oil production, both of which are critical indicators of fluid flow behaviour and recovery efficiency.

Results & Discussion

Coreflooding experiments conducted on sample core provided quantitative insights into the oil recovery performance under sequential flooding scenarios, as shown in Figure 2. The initial waterflooding phase, conducted at a constant flow rate of 0.3 cc/min as well, yielded an oil recovery of 40.00% after injecting approximately 1.28 pore volumes (PV), indicating effective displacement of mobile oil under baseline conditions. Following this, nanoflooding was implemented using a 0.1 wt.% nanosilica-based nanofluid at the same flow rate. A total of 10.17 PV was injected, resulting in a substantial incremental oil recovery of 9.78%. Notably, oil production persisted beyond 3 PV of nanofluid injection, demonstrating the extended efficacy of nanoparticle-assisted displacement mechanisms. Subsequent to nanoflooding, a secondary waterflood (WF2) was performed using brine reinjection at the same flow rate. Approximately 0.97 PV was injected until pressure equilibrium was achieved, yielding an additional oil recovery of 0.20%. The cumulative oil recovery for Core-B reached 49.98%, indicating a significant improvement over conventional waterflooding alone. Pressure monitoring throughout the experiment showed an increase from 210 psi during initial waterflooding to 208 psi during nanoflooding, and a peak of 233 psi during WF2, reflecting the dynamic flow resistance and fluid-rock interactions across the flooding stages. This indicated the permeability impairment occurred inside the core with residual retention factor (RRF) is greater than 1 (calculated at ~ 1.1). The measured NPs adsorption was 0.08 mg/gr rocks, indicating that higher adsorption led to greater recovery efficiency.

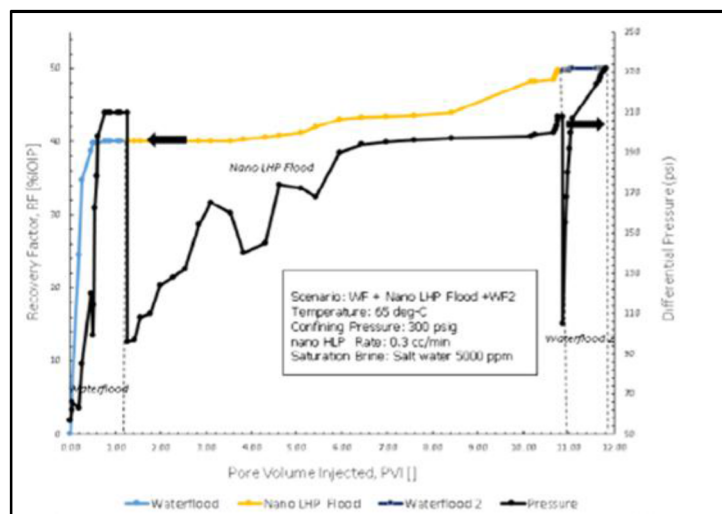


Figure 2—Coreflooding performance with differential pressure and cumulative oil recovery trends. The plot shows the potential of NPs in Carbonate core experiment with incremental oil.

The simulation results demonstrated a strong agreement with experimental observations, validating the model's predictive capability for nanofluid injection in carbonate reservoirs. As shown in Figure 3, the history matching outcomes for both differential pressure and cumulative oil recovery were within acceptable limits, confirming the model's reliability in capturing the dynamic response of the core system. Key objective functions, differential pressure (DP) and cumulative oil recovery, were effectively matched through iterative

adjustments to model parameters. Among these, nanoparticle adsorption onto the rock surface emerged as the most influential factor affecting the production profile. This finding aligns with previous work by [Hendraningrat et al. \(2023\)](#), who observed that nanoparticle adsorption and permeability impairment were primarily governed by the residual retention factor ($RRF > 1$). When nanoparticles agglomerate into larger clusters, they risk becoming trapped in narrow pore throats, leading to flow separation and sedimentation. Such sedimentation contributes to permeability reduction, with the extent of impairment closely tied to pore throat size. These insights underscore the importance of accurately modelling nanoparticle behavior, particularly adsorption dynamics, to optimize injection strategies. The validated model offers a robust framework for designing efficient nanofluid-based IOR techniques, enhancing oil recovery while mitigating formation damage in carbonate reservoirs.

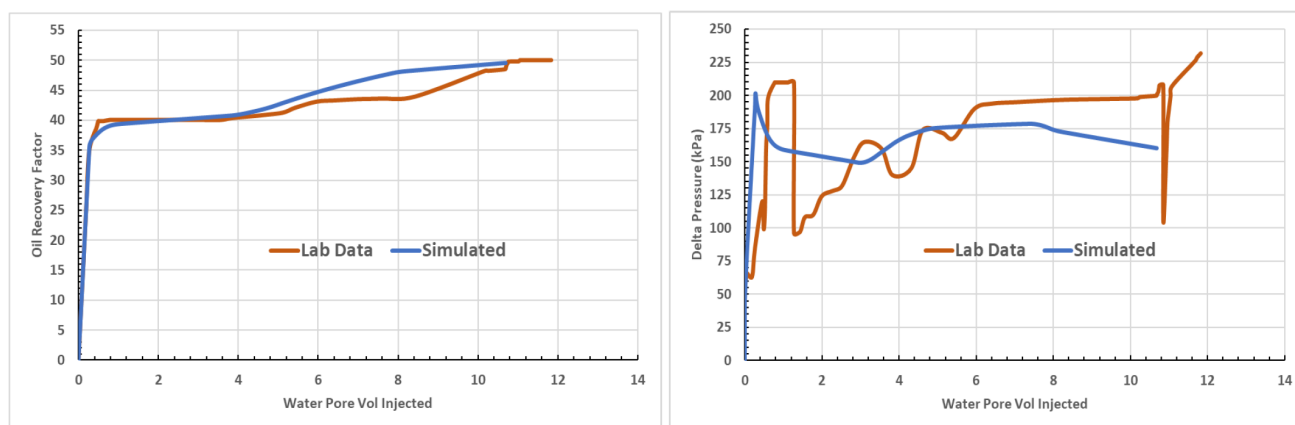


Figure 3—Model validation on Carbonate Core at reservoir pressure and temperature. The figure shows differential pressure and cumulative oil recovery trends, validating the predictive simulation model against experimental data show strong agreement.

The validated model offers a practical framework for scaling nanoparticle-assisted EOR strategies to field applications. Future work should include more coreflooding experiments prior to pilot testing in carbonate reservoirs. A preliminary cost-benefit analysis suggests that low-dosage nanoparticle injection (0.1 wt.%) can be economically viable, especially when compared to conventional chemical EOR methods. Further economic modeling is recommended to be explored with further sensitivity analysis. Silica-based nanoparticles used in this study are considered environmentally benign. However, long-term environmental safety assessments should be conducted to ensure sustainability and regulatory compliance prior to maximizing hydrocarbon recovery and ultimately contribute to the sustainable development of carbonate oil fields.

Conclusions

A comprehensive experimental and numerical modeling development was conducted to evaluate the performance of nanoparticles in carbonate rock reservoirs under reservoir conditions. The experimental coreflooding tests, which included rocks-fluid characterization and nanofluid injection, were successfully validated through a robust modeling framework. The model achieved a strong correlation with observed data, particularly in capturing the adsorption behavior of nanofluids onto carbonate surfaces as a key mechanism influencing wettability alteration and oil displacement efficiency. Adsorption was identified as a critical matching parameter, directly contributing to enhanced oil recovery. The integration of experimental insights with predictive modeling enables a deeper understanding of nanoparticle transport and interaction within complex carbonate matrices.

Future work should involve a broader experimental dataset using a larger number of core plugs to capture variability in reservoir rock properties. This will improve the reliability of pore-scale models and support more accurate upscaling to field-scale predictions. Coupling detailed core analysis with advanced numerical simulations and data-driven approaches can enhance model robustness and guide optimized reservoir management strategies. Field-scale pilot testing is essential to validate simulation outcomes and optimize nanofluid injection strategies under real reservoir conditions.

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Nomenclature

PV	= Pore Volume
DP	= Differential Pressure
EOR	= Enhanced Oil Recovery
IOR	= Improved Oil Recovery
NPs	= Nanoparticles
RF	= Recovery Factor
RRF	= Residual Resistance Factor
WF	= Waterflooding

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